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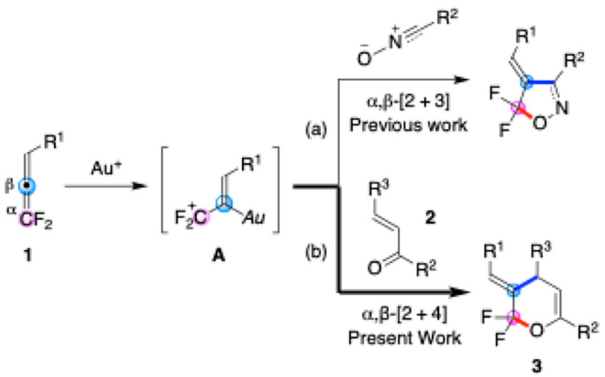
Gold(I)-catalysed [2+4] cycloaddition of 1,1-difluoroallenes with conjugated enones: synthesis of ring-difluorinated dihydro-2*H*-pyrans†

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1,1-Difluoroallenes underwent regioselective [2+4] cycloaddition with α,β-unsaturated ketones (enones) in the presence of an AuCl(IPr)–$AgSbF_6$ catalyst. β-Aurated, charge-localized difluoroallylic carbocations, formed from 1,1-difluoroallenes and the cationic Au catalyst, were stabilised by the two fluorine substituents. This facilitated α-selective nucleophilic attack by the enone oxygen, followed by six-membered ring closure, leading to the formation of ring-difluorinated dihydro-2*H*-pyrans with (*E*)-alkylidene groups in high yields.

1,1-Difluoroallenes [R(R′)C═C═CF_2], characterised by two fluorine substituents on a cumulated diene framework, have emerged as highly versatile synthetic building blocks for both cyclic and acyclic organofluorine compounds.[1,2] The unique properties of the fluorine substituents,[3] combined with the reactive nature of the allene scaffold,[4] enable pathways to valuable fluorinated targets. Over the past decade, the synthetic potential of 1,1-difluoroallenes has been extensively explored, particularly through metal-catalysed or -promoted additions and substitutions. For instance, additions involving carbon[5] and heteroatom[6] nucleophiles have facilitated the synthesis of diverse difluorovinylic and difluoroallylic compounds, while substitutions have yielded monofluoroallenes.[2c,7] Moreover, the intramolecular versions of these additions and substitutions allowed the construction of difluoromethylated[8] and fluorinated[9] naphthalene nuclei.

In contrast, the cycloadditions of 1,1-difluoroallenes have been restricted to the pioneering studies of metal-free reactions by Dolbier[2a,10] and have since remained underdeveloped, except for a $Mo(CO)_6$-catalyed reaction by Hammond.[11] In recent years, our research has focused on the metal-catalysed cycloaddition of 1,1-difluoroallenes, leading to the discovery of a gold-catalysed [2+3] cycloaddition with 1,3-dipolar compounds, such as nitrile- or imine oxides, to construct five-membered ring structures (Scheme 1a).[12] This reaction proceeds through aurated difluoroallylic cations **A** with a *Z* configuration,[6a] where the positive charge is localised owing to the unique α-carbocation stabilising effect of the fluorine substituents (Fig. 1).[3] The CF_2 cations **A** were regioselectively attacked by the oxygen[6a] of nitrile or imine oxides. The resulting adducts then underwent ring closure at the aurated carbon, leading to the formation of ring-fluorinated isoxazole derivatives (α,β-[2+3] cycloaddition).



Scheme 1 Gold-catalysed [2+3] and [2+4] cycloadditions of 1,1-difluoroallenes: past studies and recent advances.

Recognising the potential of 1,1-difluoroallene cycloadditions for constructing ring-fluorinated heterocycles, we shifted our focus towards synthesising six-membered ring-fluorinated heterocycles using α,β-unsaturated ketones in a [2+4] cycloaddition of difluoroallenes (Scheme 1b).[13,14] We hypothesised

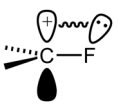


Fig. 1 α-Carbocation stabilising effect of fluorine substituent.

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that treating 1,1-difluoroallenes with enones in the presence of a gold(I) catalyst would promote the α-selective nucleophilic attack of the enone oxygen,[6a] yielding the corresponding adducts. Subsequent ring closure at the β-carbon of the enone moiety could regio- and stereoselectively generate 3-alkylidene-2,2-difluoro-3,4-dihydro-2*H*-pyrans **3** (α,β-[2+4] cycloaddition). Notably, difluorodihydro-2*H*-pyrans have applications as (i) liquid crystal materials (Fig. 2)[15] and as antiulcer agents.[16,17] (ii) Products **3** would act as key intermediates with an allylic difluoride moiety in the construction of ring-fluorinated pyran systems *via* S_N2'-type reactions.[18]

We initiated our investigation using difluoroallene **1a** and *trans*-benzalacetone (**2a**, Fig. 3) as model substrates (Table 1). First, using 2 mol% of AuCl and $AuCl_3$ as Au(I) and Au(III) catalysts, respectively, produced 2,2-difluoro-3,4-dihydro-2*H*-pyran **3a** in 6% and 5% yields at the reflux temperature of 1,2-dichloroethane, both in an *E*-selective manner (entries 1 and 2). When a cationic $AuCl(PPh_3)$–$AgSbF_6$ (1 : 1) catalyst was used, the starting material **1a** was consumed, yielding **3a** in 10% yield (entry 3). Notably, the yield of **3a** considerably improved upon using an AuCl(IPr)–$AgSbF_6$ catalyst in the presence of molecular sieves 4A [IPr = 1,3-bis(2,6-diisopropylphenyl)imidazol-2-ylidene; entry 4], where the desired **3a** was obtained in 86% yield at room temperature.[19] Of note, the [2+2] cycloadduct **4a** was not observed during the optimisation.

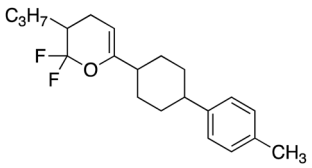


Fig. 2 Beneficial ring-fluorinated dihydro-2*H*-pyran.

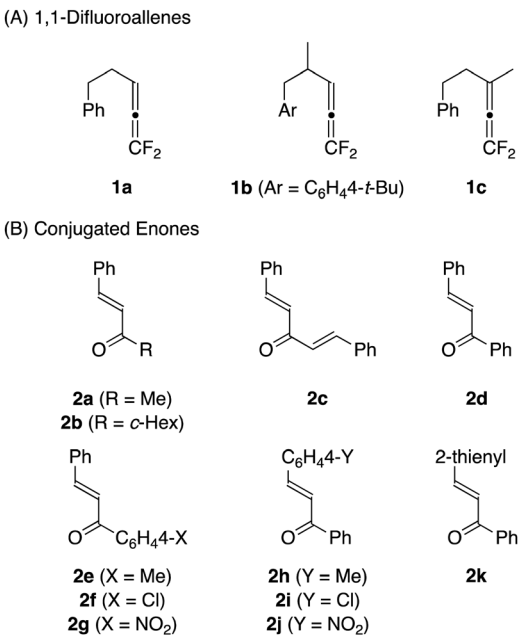
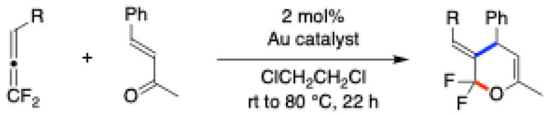


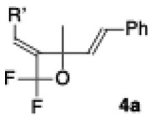
Fig. 3 List of substrates.

Table 1 Optimisation of metal catalysts in [2+4] cycloaddition of 1,1-difluoroallene **1a** with enone **2a**



Entry	Catalyst	**3**[a] (%)	**1a**[a,b] (%)
1	AuCl	6	34
2	$AuCl_3$	5	26
3	$AuCl(PPh_3)$–$AgSbF_6$ (1 : 1)	10	nd
4[c,d]	AuCl(IPr)–$AgSbF_6$ (1 : 1)	86	nd

R = CH_2CH_2Ph. nd = not detected by ^{19}F NMR. [a] ^{19}F NMR yield based on an internal standard $PhCF_3$. [b] Recovery. [c] rt, 1 h. [d] The reaction was performed in the presence of MS 4A.



A survey of solvents, using chalcone (**2d**) as the enone, identified 1,2-dichloroethane as the most effective solvent (Table 2). When **1a** was reacted with **2d** in the presence of an AuCl(IPr)–$AgSbF_6$ catalyst in toluene, **3f** was obtained in 3% yield (entry 1). Performing the reaction in hexane increased the yield of **3f** to 52% (entry 2). Although the reaction reached completion in dichloromethane, it had poor reproducibility (entry 3), whereas in 1,2-dichloroethane, it achieved a quantitative yield (entry 4). When the reaction was performed in ether, THF and DMF, the starting difluoroallene **1a** was recovered in yields of 24–72% (entries 5–7).

The [2+4] cycloaddition of difluoroallenes with enones **2** enabled the synthesis of (*E*)-3-alkylidene-2,2-difluorodihydro-2*H*-pyrans **3** as single isomers (Table 3). Starting with the benchmark allene **1a**, the reaction produced **3a** in 90% isolated yield. Similarly, difluoroallenes **1b** and **1c** reacted with enones **2a** and **2d**, yielding the corresponding ring-fluorinated difluoro(dihydro)pyrans **3b** and **3c** in 83% and 97% yields, respectively. Enones **2b** and **2c**, featuring a secondary alkyl group and an alkenyl group as R^3, respectively, underwent reactions with difluoroallene **1a** to afford **3d** and **3e** in

Table 2 Optimisation of solvents in [2+4] cycloaddition of 1,1-difluoroallene **1a** with enone **2d**

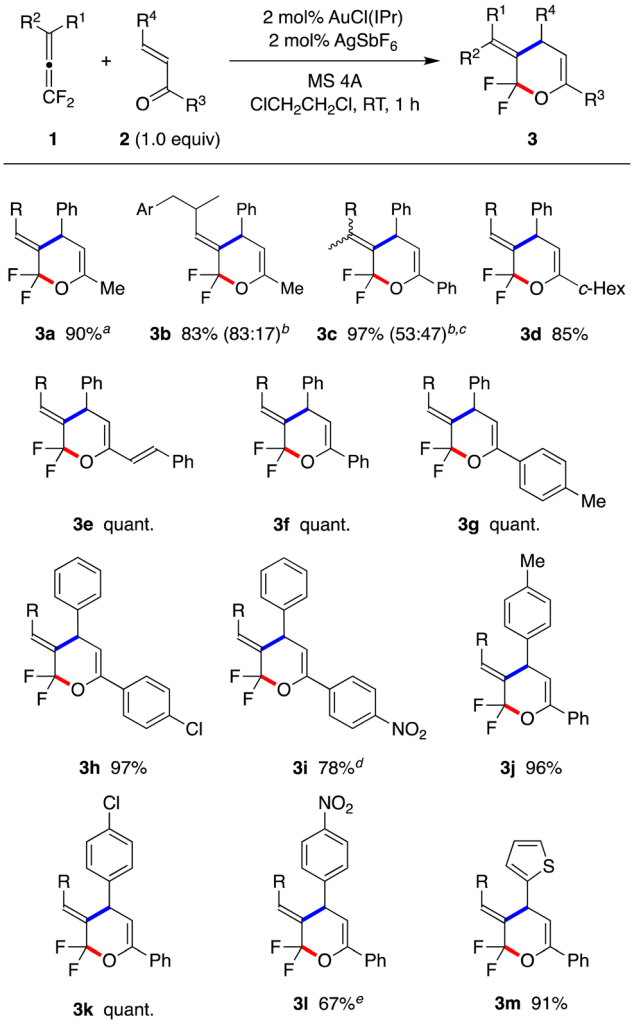
Entry	Solvent	Conditions	**3f**[a] (%)	**1a**[a,b] (%)
1	Toluene	rt to 80 °C, 6 h	3	81
2	Hexane	rt to reflux, 19 h	52	28
3	CH_2Cl_2	rt, 2 h	3–90	nd
4	$ClCH_2CH_2Cl$	rt, 1 h	Quant.	nd
5	Et_2O	rt, 20 h	3	72
6	THF	rt to reflux, 20 h	nd	39
7	DMF	rt to 80 °C, 19 h	nd	24

nd = not detected by ^{19}F NMR. Conditions: **1a** (1.0 equiv.), **2d** (1.0 equiv.), 2 mol% $AuCl(PPh_3)$, 2 mol% $AgSbF_6$, MS 4A (530 mg/**1a** 1 mmol). [a] ^{19}F NMR yield based on an internal standard $PhCF_3$. [b] Recovery.

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Table 3 Synthesis of (*E*)-3-alkylidene-2,2-difluorodihydro-2*H*-pyrans **3**[a]

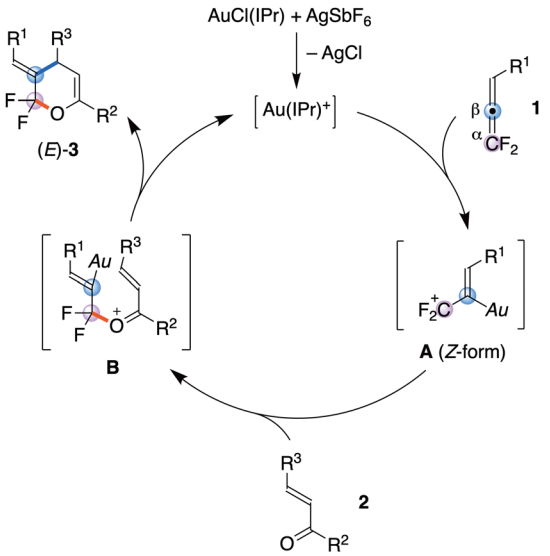


Isolated yield. R = CH_2CH_2Ph. Ar = $C_6H_4$4-*t*-Bu. [a] Table 1, entry 4. [b] Diastereomer ratio in parentheses. [c] 5 mol% AuCl(IPr), 5 mol% $AgSbF_6$. [d] CH_2Cl_2 was used as a solvent. [e] RT to 80 °C, 5 h.

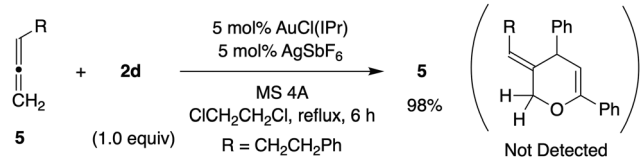
85% and quantitative yields. In addition to the alkylated and alkenylated **2a–c**, arylated enones **2d–g**, including those with electron-donating and electron-withdrawing groups on the aromatic rings (R^3 = Ph, $C_6H_4$4-Me, $C_6H_4$4-Cl, $C_6H_4$4-NO_2) reacted with **1a** to produce **3f–i** in yields ranging from 78% to quantitative.

Additionally, conjugated enones with various substituents at the β-position (R^4, **2h–j**) reacted with difluoroallene **1a** to yield the corresponding products **3j–l** in 67% to quantitative yields as single diastereomers regarding the *exo*-alkylidene moiety.[20] A hetaryl-substituted difluoro(dihydro)pyran, such as **3m**, was synthesised from the corresponding 2-thienyl enone **2k** in 91% yield.

The proposed catalytic cycle for the α,β-selective [2+4] cycloaddition is depicted in Scheme 2. Initially, difluoroallenes **1** are activated by the AuCl(IPr)–$AgSbF_6$ catalyst, generating the *Z*-form of the aurated, charge-localised difluoroallylic cations **A**.[6a] As expected from our former results,[6a] nucleophilic attack by the



Scheme 2 Plausible catalytic cycle of α,β-selective [2+4] cycloaddition of 1,1-difluoroallenes **1** with enones **2**.



Scheme 3 Control experiment with nonfluorinated allene **5**.

oxygen atom of enones **2** occurs regioselectively at the carbon α to the fluorine substituents, forming intermediate adducts **B**. Finally, ring closure of **B** proceeds to yield the α,β-[2+4] cycloadducts (*E*)-**3**, maintaining the configuration of the allylic cations **A**. The cationic Au catalyst is regenerated to complete the catalytic cycle.

Nonfluorinated allene **5** did not react with chalcone (**2d**) in the presence of the same AuCl(IPr)–$AgSbF_6$ catalyst, leading to a 98% recovery of the starting **5** even at the reflux temperature of 1,2-dichloroethane (Scheme 3). Thus, the α-carbocation stabilising effect of the fluorine substituents contributed greatly to the generation of the key allylic cations, *i.e.* **A**, thereby facilitating the synthesis of the ring-difluorinated dihydro-2*H*-pyrans.

In summary, the [2+4] cycloaddition of 1,1-difluoroallenes and conjugated enones successfully yielded ring-difluorinated oxygen heterocycles. In the presence of a cationic gold(I) catalyst, the reaction proceeded with regio- and stereoselectivities for both the difluoroallenes and enones, resulting in the formation of six-membered products, (*E*)-3-alkylidene-2,2-difluoro-3,4-dihydro-2*H*-pyrans, in good to excellent yields.

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Science and Technology Foundation. We acknowledge TOSOH FINECHEM CORPORATION for the generous gift of CF_3CH_2I.

Data availability

The data supporting this article have been included as part of the ESI.†

Conflicts of interest

There are no conflicts to declare.

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19 MS 4A was used to prevent the hydrolysis of 1,1-difluoroallenes **1** and difluoro(dihydro)pyrans **3**. For example, when the reaction was performed in the absence of MS 4A, the yield of **3a** decreased to 77%, based on ^{19}F NMR analysis.
20 Homo- and heteronuclear NOE experiments indicate an *E* geometry for the alkylidene moiety in **3l** (and those in other products **3**), which was also observed in our previous systems in ref. 6a and 12.

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